

## SAFETY DATA SHEET

Revision Date 01-Apr-2024

Revision Number 3

### 1. Identification

**Product Name** **Samarium(III) isopropoxide, 99% (REO), 5% w/v in toluene/isopropanol**

**Cat No. :** **41311**

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### Emergency Telephone Number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                      |            |
|------------------------------------------------------|------------|
| Flammable liquids                                    | Category 2 |
| Skin Corrosion/Irritation                            | Category 2 |
| Serious Eye Damage/Eye Irritation                    | Category 2 |
| Reproductive Toxicity                                | Category 2 |
| Specific target organ toxicity (single exposure)     | Category 3 |
| Target Organs - Central nervous system (CNS).        |            |
| Specific target organ toxicity - (repeated exposure) | Category 2 |
| Target Organs - Neurological effects, Eyes, Ears.    |            |
| Aspiration Toxicity                                  | Category 1 |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Highly flammable liquid and vapor  
May be fatal if swallowed and enters airways  
Causes skin irritation  
Causes serious eye irritation  
May cause drowsiness or dizziness  
Suspected of damaging the unborn child  
May cause damage to organs through prolonged or repeated exposure



**Precautionary Statements**

**Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Wash face, hands and any exposed skin thoroughly after handling  
Wear eye/face protection  
Do not breathe dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Keep cool

**Response**

IF exposed or concerned: Get medical attention/advice

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

If skin irritation occurs: Get medical advice/attention  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower  
Wash contaminated clothing before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Ingestion**

IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician  
Do NOT induce vomiting

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

WARNING. Reproductive Harm - <https://www.p65warnings.ca.gov/>.

### 3. Composition/Information on Ingredients

| Component                              | CAS No    | Weight % |
|----------------------------------------|-----------|----------|
| Toluene                                | 108-88-3  | 47.50    |
| Isopropyl alcohol                      | 67-63-0   | 47.50    |
| Samarium(III) isopropoxide in ampoules | 3504-40-3 | 5.00     |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                     |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                              |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                     |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                   |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).                                   |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward. |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting                                                        |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                               |

### 5. Fire-fighting measures

|                                         |                                                   |
|-----------------------------------------|---------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                          |
| <b>Flash Point</b>                      | 4 °C / 39.2 °F                                    |
| <b>Method -</b>                         | No information available                          |
| <b>Autoignition Temperature</b>         | No information available                          |
| <b>Explosion Limits</b>                 |                                                   |
| <b>Upper</b>                            | No data available                                 |
| <b>Lower</b>                            | No data available                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                          |

#### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Samarium oxide.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
3

**Flammability**  
3

**Instability**  
0

**Physical hazards**  
-

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.              |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Strong bases.                                                                                                                                                                                                                                                                                      |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component         | ACGIH TLV                     | OSHA PEL                                                                                                                                                                          | NIOSH                                                                                                         | Mexico OEL (TWA)              |
|-------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------|
| Toluene           | TWA: 20 ppm                   | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 375 mg/m <sup>3</sup><br>Ceiling: 300 ppm<br>(Vacated) STEL: 150 ppm<br>(Vacated) STEL: 560 mg/m <sup>3</sup><br>TWA: 200 ppm            | IDLH: 500 ppm<br>TWA: 100 ppm<br>TWA: 375 mg/m <sup>3</sup><br>STEL: 150 ppm<br>STEL: 560 mg/m <sup>3</sup>   | TWA: 20 ppm                   |
| Isopropyl alcohol | TWA: 200 ppm<br>STEL: 400 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 980 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 1225 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup> | IDLH: 2000 ppm<br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1225 mg/m <sup>3</sup> | TWA: 200 ppm<br>STEL: 400 ppm |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists  
 OSHA - Occupational Safety and Health Administration  
 NIOSH: NIOSH - National Institute for Occupational Safety and Health

|                             |                                                                                                                                                                                                                  |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering Measures</b> | Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Personal Protective Equipment

|                                 |                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eyeface Protection</b>       | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                   |

|                                 |                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>   | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Recommended Filter type:</b> | low boiling organic solvent. Type AX. Brown. conforming to EN371. or. Organic gases and vapours filter. Type A. Brown. conforming to EN14387.                                                                                                     |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b>Physical State</b>                         | Liquid                   |
| <b>Appearance</b>                             | No information available |
| <b>Odor</b>                                   | No information available |
| <b>Odor Threshold</b>                         | No information available |
| <b>pH</b>                                     | No information available |
| <b>Melting Point/Range</b>                    | No data available        |
| <b>Boiling Point/Range</b>                    | No information available |
| <b>Flash Point</b>                            | 4 °C / 39.2 °F           |
| <b>Evaporation Rate</b>                       | No information available |
| <b>Flammability (solid,gas)</b>               | Not applicable           |
| <b>Flammability or explosive limits</b>       |                          |
| Upper                                         | No data available        |
| Lower                                         | No data available        |
| <b>Vapor Pressure</b>                         | 23 hPa @ 20 °C           |
| <b>Vapor Density</b>                          | No information available |
| <b>Specific Gravity</b>                       | No information available |
| <b>Solubility</b>                             | No information available |
| <b>Partition coefficient; n-octanol/water</b> | No data available        |
| <b>Autoignition Temperature</b>               | No information available |
| <b>Decomposition Temperature</b>              | No information available |
| <b>Viscosity</b>                              | No information available |
| <b>Molecular Formula</b>                      | C9 H21 O3 Sm             |
| <b>Molecular Weight</b>                       | 327.50                   |

## 10. Stability and reactivity

|                                         |                                                                         |
|-----------------------------------------|-------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                              |
| <b>Stability</b>                        | Moisture sensitive.                                                     |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition.       |
| <b>Incompatible Materials</b>           | Strong bases                                                            |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Samarium oxide |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                |
| <b>Hazardous Reactions</b>              | None under normal processing.                                           |

## 11. Toxicological information

### Acute Toxicity

|                              |                                                                               |
|------------------------------|-------------------------------------------------------------------------------|
| <b>Product Information</b>   |                                                                               |
| <b>Oral LD50</b>             | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| <b>Dermal LD50</b>           | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| <b>Vapor LC50</b>            | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |
| <b>Component Information</b> |                                                                               |

| Component         | LD50 Oral                                  | LD50 Dermal                   | LC50 Inhalation       |
|-------------------|--------------------------------------------|-------------------------------|-----------------------|
| Toluene           | > 5000 mg/kg ( Rat )                       | LD50 = 12000 mg/kg ( Rabbit ) | 26700 ppm ( Rat ) 1 h |
| Isopropyl alcohol | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat )           | 72.6 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                              | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------------|-----------|------------|------------|------------|------------|------------|
| Toluene                                | 108-88-3  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Isopropyl alcohol                      | 67-63-0   | Not listed | Not listed | Not listed | Not listed | Not listed |
| Samarium(III) isopropoxide in ampoules | 3504-40-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Central nervous system (CNS)

**STOT - repeated exposure** Neurological effects Eyes Ears

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

The product contains following substances which are hazardous for the environment. Contains a substance which is: Toxic to aquatic organisms. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component         | Freshwater Algae                                                                                                           | Freshwater Fish                                                                                                                         | Microtox                                           | Water Flea                                                                                   |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------|
| Toluene           | EC50: = 12.5 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: > 433 mg/L, 96h (Pseudokirchneriella subcapitata) | 50-70 mg/L LC50 96 h<br>5-7 mg/L LC50 96 h<br>15-19 mg/L LC50 96 h<br>28 mg/L LC50 96 h<br>12 mg/L LC50 96 h                            | EC50 = 19.7 mg/L 30 min                            | EC50: = 11.5 mg/L, 48h (Daphnia magna)<br>EC50: 5.46 - 9.83 mg/L, 48h Static (Daphnia magna) |
| Isopropyl alcohol | EC50: > 1000 mg/L, 72h (Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 96h (Desmodesmus subspicatus)                       | LC50: = 9640 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h (Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h | = 35390 mg/L EC50 Photobacterium phosphoreum 5 min | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h                                              |

|  |  |                                                                      |  |  |
|--|--|----------------------------------------------------------------------|--|--|
|  |  | static (Pimephales promelas)<br>LC50: = 10000000 µg/L, 96h (Daphnia) |  |  |
|--|--|----------------------------------------------------------------------|--|--|

**Persistence and Degradability** Immiscible with water May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component         | log Pow |
|-------------------|---------|
| Toluene           | 2.73    |
| Isopropyl alcohol | 0.05    |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component          | RCRA - U Series Wastes | RCRA - P Series Wastes |
|--------------------|------------------------|------------------------|
| Toluene - 108-88-3 | U220                   | -                      |

### 14. Transport information

#### DOT

UN-No UN3274  
 Proper Shipping Name ALCOHOLATES SOLUTION, N.O.S.  
 Technical Name (Samarium(III) isopropoxide, TOLUENE)  
 Hazard Class 3  
 Subsidiary Hazard Class 8  
 Packing Group II

#### TDG

UN-No UN3274  
 Proper Shipping Name ALCOHOLATES SOLUTION, N.O.S.  
 Hazard Class 3  
 Subsidiary Hazard Class 8  
 Packing Group II

#### IATA

UN-No UN3274  
 Proper Shipping Name ALCOHOLATES SOLUTION, N.O.S.  
 Hazard Class 3  
 Subsidiary Hazard Class 8  
 Packing Group II

#### IMDG/IMO

UN-No UN3274  
 Proper Shipping Name ALCOHOLATES SOLUTION, N.O.S.  
 Hazard Class 3  
 Subsidiary Hazard Class 8  
 Packing Group II

### 15. Regulatory information

#### United States of America Inventory

| Component                     | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-------------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Toluene                       | 108-88-3  | X    | ACTIVE                                        | -                           |
| Isopropyl alcohol             | 67-63-0   | X    | ACTIVE                                        | -                           |
| Samarium(III) isopropoxide in | 3504-40-3 | -    | -                                             | -                           |

**Samarium(III) isopropoxide, 99% (REO), 5% w/v in toluene/isopropanol**

Revision Date 01-Apr-2024

|          |  |  |  |  |
|----------|--|--|--|--|
| ampoules |  |  |  |  |
|----------|--|--|--|--|

**Legend:****TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                              | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|----------------------------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Toluene                                | 108-88-3  | X   | -   | 203-625-9 | X     | X    | X    | X    | X     | KE-33936 |
| Isopropyl alcohol                      | 67-63-0   | X   | -   | 200-661-7 | X     | X    | X    | X    | X     | KE-29363 |
| Samarium(III) isopropoxide in ampoules | 3504-40-3 | -   | -   | -         | -     | -    |      | -    | -     | -        |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component         | CAS No   | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|-------------------|----------|----------|-------------------------------|---------------------------------|
| Toluene           | 108-88-3 | 47.50    | 1.0 %                         | -                               |
| Isopropyl alcohol | 67-63-0  | 47.50    | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|-----------|----------------------------|-----------------------------|------------------------|---------------------------|
| Toluene   | X                          | 1000 lb                     | X                      | X                         |

**Clean Air Act**

| Component | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|-----------|-----------|-------------------------|-------------------------|
| Toluene   | X         |                         | -                       |

**OSHA** - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances | SARA Reportable Quantity (RQ) |
|-----------|--------------------------|---------------------------------------|-------------------------------|
|-----------|--------------------------|---------------------------------------|-------------------------------|



|         |         | <b>RQs</b> |                   |
|---------|---------|------------|-------------------|
| Toluene | 1000 lb | -          | 1000 lb<br>454 kg |

**California Proposition 65** This product contains the following Proposition 65 chemicals.

| <b>Component</b> | <b>CAS No</b> | <b>California Prop. 65</b> | <b>Prop 65 NSRL</b> | <b>Category</b> |
|------------------|---------------|----------------------------|---------------------|-----------------|
| Toluene          | 108-88-3      | Developmental              | -                   | Developmental   |

**U.S. State Right-to-Know Regulations**

| <b>Component</b>  | <b>Massachusetts</b> | <b>New Jersey</b> | <b>Pennsylvania</b> | <b>Illinois</b> | <b>Rhode Island</b> |
|-------------------|----------------------|-------------------|---------------------|-----------------|---------------------|
| Toluene           | X                    | X                 | X                   | X               | X                   |
| Isopropyl alcohol | X                    | X                 | X                   | -               | X                   |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH**

| <b>Component</b>                       | <b>CAS No</b> | <b>REACH (1907/2006) - Annex XIV - Substances Subject to Authorization</b> | <b>REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances</b>                                                     | <b>REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)</b> |
|----------------------------------------|---------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Toluene                                | 108-88-3      | -                                                                          | Use restricted. See item 48.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                            |
| Isopropyl alcohol                      | 67-63-0       | -                                                                          | Use restricted. See item 75.<br>(see link for restriction details)                                                                       | -                                                                                                            |
| Samarium(III) isopropoxide in ampoules | 3504-40-3     | -                                                                          | -                                                                                                                                        | -                                                                                                            |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| <b>Component</b>              | <b>CAS No</b> | <b>OECD HPV</b> | <b>Persistent Organic Pollutant</b> | <b>Ozone Depletion Potential</b> | <b>Restriction of Hazardous Substances (RoHS)</b> |
|-------------------------------|---------------|-----------------|-------------------------------------|----------------------------------|---------------------------------------------------|
| Toluene                       | 108-88-3      | Listed          | Not applicable                      | Not applicable                   | Not applicable                                    |
| Isopropyl alcohol             | 67-63-0       | Listed          | Not applicable                      | Not applicable                   | Not applicable                                    |
| Samarium(III) isopropoxide in | 3504-40-3     | Not applicable  | Not applicable                      | Not applicable                   | Not applicable                                    |

|          |  |  |  |  |  |
|----------|--|--|--|--|--|
| ampoules |  |  |  |  |  |
|----------|--|--|--|--|--|

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component                              | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Toluene                                | 108-88-3  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |
| Isopropyl alcohol                      | 67-63-0   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |
| Samarium(III) isopropoxide in ampoules | 3504-40-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

**Prepared By**

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date**

01-Apr-2024

**Print Date**

01-Apr-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**